

Demo: Loading Macros via include-in-header

This page demonstrates the `include-in-header` strategy for loading macros. The YAML front matter of this file includes:

```
format:
  html:
    include-in-header:
      - macros-header.html    # MathJax macro config script
  pdf:
    include-in-header:
      - macros.qmd            # raw LaTeX goes directly into preamble
  revealjs:
    output-file: demo-include-in-header-slides.html
    include-in-header:
      - macros-header.html    # loader that reads macros.qmd
docx: default
```

Note (HTML/RevealJS): `macros-header.html` is a small loader script (not a second macro-definition file). It reads `macros.qmd`, converts macro definitions to a MathJax config object, and sets `MathJax.tex.macros` before typesetting. For RevealJS, it also forces the built-in Reveal math plugin to use MathJax 3. [View the RevealJS slides output.](#)

Note (PDF): For PDF, `macros.qmd` can be included directly since its raw LaTeX goes into the document preamble.

Note: This demo uses `macros-header.html` / `macros.qmd` because both files are at the root of this repository. When using this repository as a Git submodule (cloned into a `macros/` subdirectory), replace with `macros/macros-header.html` and `macros/macros.qmd`.

The same configuration can be placed in a shared `_quarto.yml` to apply project-wide:

```

format:
  html:
    include-in-header:
      - macros/macros-header.html
  pdf:
    include-in-header:
      - macros/macros.qmd
  revealjs:
    include-in-header:
      - macros/macros-header.html

```

This inserts the macro definitions for each output format:

- **PDF:** `macros.qmd` goes into the LaTeX preamble via `include-in-header` — all macros work, including `\providecommand`.
- **HTML:** `macros-header.html` is inserted via `include-in-header` and loads macros from `macros.qmd` into `MathJax.tex.macros`.
- **RevealJS:** `macros-header.html` does the same loader step and switches the Reveal math plugin to MathJax 3, so slide math uses the same `macros.qmd` source.
- **docx:** Word documents do not use a LaTeX preamble or MathJax header, so `include-in-header` has no effect on math rendering for docx. The `docx: default` entry above produces docx output without macro support.

Example math

The table below uses macros defined with `\def` in `macros.qmd`, which work in all output formats that support MathJax or LaTeX:

Description	Code	Output
Greek: alpha	<code>\$(a)\$</code>	α
Greek: beta	<code>\$(b)\$</code>	β
Greek: gamma	<code>\$(g)\$</code>	γ
Greek: lambda	<code>\$(l)\$</code>	λ
Greek: sigma	<code>\$(s)\$</code>	σ
Log-likelihood	<code>\$(llik)\$</code>	ℓ
Independent	<code>\$(X \ind Y)\$</code>	$X \perp\!\!\!\perp Y$
IID distributed	<code>\$(X_i \siid \Normal(\mu, \ss))\$</code>	$X_i \sim_{\text{iid}} N(\mu,)$
Vector: mu	<code>\$(\vm)\$</code>	$\vec{\mu}$
Hat beta	<code>\$(\hb)\$</code>	$\hat{\beta}$